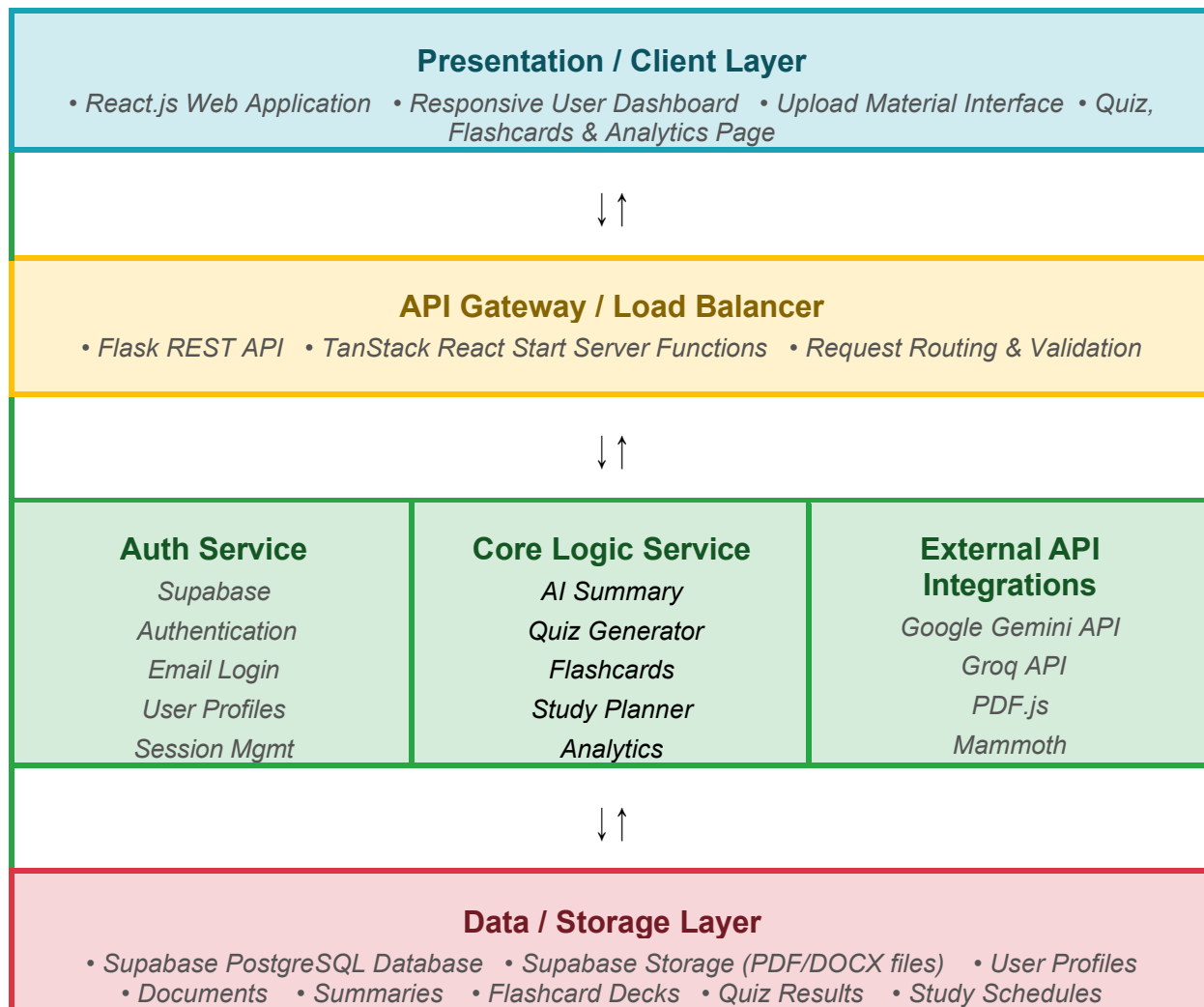


Solution Architecture

Date	15 July 2026
Team ID	XXXXXX
Project Name	StudyAI – Smart AI educate Companion & Quiz Generator
Maximum Marks	5 Marks

Solution Architecture Diagram:



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
<i>[Presentation Layer]</i>	<i>Provides the user interface where students can register, log in, upload study materials, generate summaries, flashcards, quizzes, and view analytics dashboards.</i>	<i>React.js, TypeScript, Tailwind CSS, HTML5, CSS3, Vite</i>
<i>[API Gateway]</i>	<i>Receives requests from the frontend, validates them, routes them to appropriate services, and returns responses to the client.</i>	<i>Flask REST API, TanStack Server Functions</i>
<i>[Microservices]</i>	<i>Handles core business logic such as AI-powered summary generation, quiz generation, flashcard creation, study plan generation, file processing, and analytics computation.</i>	<i>Python Flask, Google Gemini API/Groq API, PDF.js, Mammoth</i>
<i>[Database]</i>	<i>Stores user accounts, uploaded documents, summaries, flashcards, quiz results, study schedules, and analytics data securely.</i>	<i>Supabase PostgreSQL, Supabase Storage</i>